# US Patent & Trademark Office
# Patent Public Search | Text View


| | |
|---|---|
| United States Patent Application Publication | 20250287352 |
| Kind Code | A1 |
| Publication Date | September 11, 2025 |
| Inventor(s) | Kuravangi-Thammaiah; Shanthala et al. |


## SYSTEMS AND METHODS FOR SUBSCRIPTION-BASED PRIORITY ACCESS AND SERVICES IN A WIRELESS NETWORK


### Abstract

A system may maintain subscription information associated with a plurality of User Equipment (“UEs”), which may include priority level information, such as one or more access identities, associating a UE with one or more respective priority levels. The system may receive, from a network function (“NF”) of a wireless network, a request for priority level information associated with the UE, and may identify a particular set of priority levels indicated by the subscription information as being associated with the UE. The system may output the identified particular set of priority levels to the NF. The NF may maintain policy information associating the particular set of priority levels with one or more network parameters and may implement the one or more network parameters when providing service associated with the UE. The system may include a Unified Data Management function (“UDM”) or a Unified Data Repository (“UDR”) of the wireless network.



**Inventors:** **Kuravangi-Thammaiah; Shanthala (Keller, TX), Habib; Ambreen (Golden, CO)**

**Applicant:** **Verizon Patent and Licensing Inc.** (Basking Ridge, NJ)



**Family ID:** **96949825**

**Assignee:** **Verizon Patent and Licensing Inc. (Basking Ridge, NJ)**

**Appl. No.:** **18/598677**

**Filed:** **March 07, 2024**

### Publication Classification

**Int. Cl.:** **H04W68/02** (20090101); **H04W8/20** (20090101); **H04W28/02** (20090101)

**U.S. Cl.:**

CPC **H04W68/02** (20130101); **H04W8/20** (20130101); **H04W28/0268** (20130101);

## Background/Summary

BACKGROUND

[0001] Wireless networks provide wireless connectivity to User Equipment ("UEs"), such as mobile telephones, tablets, Internet of Things ("IoT") devices, Machine-to-Machine ("M2M") devices, or the like. Different UEs or users thereof may be associated with different categories or groups, such as "first responder," "enterprise user," "streaming device," etc. These different categories or groups may be indicative of different levels services provided by a wireless network, such as different Quality of Service ("QoS") parameters, different queuing parameters, different load balancing parameters, or the like.

---

## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. **1** illustrates an example overview of one or more embodiments described herein;

[0003] FIG. **2** illustrates example UE priority level information and/or policies, in accordance with some embodiments;

[0004] FIG. **3** illustrates an example of providing subscription-based priority access information, associated with one or more UEs, to one or more network functions ("NFs") of a wireless network, in accordance with some embodiments;

[0005] FIG. **4** illustrates an example of automatically providing updated subscription-based priority access information to one or more NFs of a wireless network, in accordance with some embodiments;

[0006] FIG. **5** illustrates example NFs that may utilize UE priority level information and/or policies when providing network connectivity to one or more UEs, in accordance with some embodiments;

[0007] FIG. **6** illustrates an example signal flow for providing subscription-based priority access to a UE in a wireless network, in accordance with some embodiments;

[0008] FIG. **7** illustrates an example process for providing subscription-based priority access to a UE in a wireless network, in accordance with some embodiments;

[0009] FIGS. **8** and **9** illustrate example environments in which one or more embodiments, described herein, may be implemented;

[0010] FIG. **10** illustrates an example arrangement of a radio access network ("RAN"), in accordance with some embodiments; and

[0011] FIG. **11** illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0012] The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

[0013] Wireless networks may implement mechanisms by which different levels of access, QoS parameters, or the like may be provided to UEs that receive wireless connectivity from such wireless networks. Such mechanisms may include implementing network slices, Data Network Name ("DNNs"), access lists, and/or other suitable mechanisms that are associated with different sets of parameters. Embodiments described herein provide for a granular, per-UE subscription-based mechanism for indicating different priority levels with which one or more UEs are associated. As discussed herein, the different priority levels may be used by the wireless network (e.g., by a RAN of the wireless network and/or a core of the wireless network) to provide different QoS parameters, access parameters, etc. to different UEs.

[0014] As also discussed herein, UE priority level information may be maintained by one or more network elements that are configured to maintain UE subscription information, such as a Unified Data Management function ("UDM") and/or a Unified Data Repository ("UDR"). Maintaining such information by such network elements may allow for other NFs of the wireless network to obtain priority level information on a per-UE basis, and to accordingly provide service to UEs based on respective priority levels. In some embodiments, the NFs may use Service-Based Interfaces ("SBIs") to obtain the UE priority level information from the UDM and/or the UDR (herein referred to as "UDM/UDR" for brevity). Further, NFs of the wireless network may be automatically updated when subscription information associated with a particular UE (e.g., including UE priority level information) is updated at the UDM/UDR (e.g., by an administrator or operator of the wireless network).

[0015] For example, as shown in FIG. **1**, RAN **101** and/or core network **103** (e.g., which may be associated with the same wireless network) may maintain UE priority level information/policies **105**. As discussed above, UE priority level information may be maintained on a per-UE basis. In this example, assume that UE priority level information/policies **105** include UE priority level information for UEs **107-1**, **107-2**, and **107-3**. For example, UE priority level information/policies **105** may indicate that UE **107-1** is associated with (e.g., authorized for) a first priority level or set of priority levels, that UE **107-2** is associated with a second priority level or set of priority levels, and that UE **107-3** is associated with a third priority level or set of priority levels.

[0016] FIG. **2** illustrates example UE priority level information/policies **105** that may be maintained by one or more elements of RAN **101** and/or core network **103**. In this example, UE priority level information/policies **105** may include priority level policies (e.g., as represented in data structure **201**) and/or UE priority level information (e.g., as represented in data structure **203**). In some embodiments, UE priority level information/policies **105** may indicate respective priority levels using access identities or other suitable identifiers. For example, a first access identity (e.g., access identity "11") may be associated with a first priority level, a second access identity (e.g., access identity "12") may be associated with a second priority level, a third access identity (e.g., access identity "13") may be associated with a third priority level, a fourth access identity (e.g., access identity "14") may be associated with a fourth priority level, and a fifth access identity (e.g., access identity "15") may be associated with a fifth priority level. In some embodiments, additional, fewer, and/or different access identities may be associated with different priority levels. In some embodiments, other priority level identifiers may be used in addition to, or in lieu of, access identities.

[0017] As shown, data structure **201** may include information associating particular access identities with particular sets of QoS parameters. In this example, QoS parameters may refer to different network slices (e.g., Network Slice Selection Assistance Information ("NSSAI") values or other suitable network slice identifiers), Data Network Names ("DNNs"), 5G QOS Identifiers ("5QIs"), or other suitable values. As shown, a particular access identity may be associated with one set of QoS parameters or multiple sets of QoS parameters. As discussed below, different network elements may use priority level policies (e.g., data structure **201**) when handling, forwarding, queuing, processing, etc. traffic sent to or received from UEs **107** that are associated with particular access identities.

[0018] Additionally, or alternatively, in some embodiments, other types of policies or QoS parameters may be associated with particular access identities or other priority level identifiers. For example, in some embodiments, the priority level policies may include RAN paging profiles and/or other RAN-specific parameters. For example, a particular RAN paging profile may indicate how often a given UE **107** should be paged, located, etc. by one or more elements of RAN **101**, such as in situations where downlink data is to be sent to such UE **107** (e.g., based on receiving a downlink data notification from core network **103**). For example, a first paging profile may indicate that a given UE **107** should be notified as soon as possible when a downlink data notification is received, while a second paging profile may indicate that a given UE **107** should not be paged more than a maximum threshold frequency (e.g., not more than once per minute, not more than once per 30 seconds, etc.). As another example, the RAN-specific parameters may include RAN radio resource allocation parameters, queuing parameters, 5G QOS Identifier ("5QI") values, and/or other suitable RAN-specific parameters.

[0019] Examples herein are discussed in the context of QoS parameters. In practice, similar concepts may apply for any suitable type of network parameter that is used by network elements to route, forward, process, filter, handle, etc. traffic sent to or sent by one or more UEs **107**, in accordance with identifying respective priority levels associated with such UEs **107**.

[0020] In some embodiments, different elements of RAN **101** and/or core network **103** may maintain different portions or subsets of priority level policy information (e.g., as represented by data structure **201**). For example, one or more elements of RAN **101** may maintain information associating particular access identities with respective paging profiles, while one or more elements of core network **103** may maintain information associating particular access identities with respective QoS parameters but may not maintain information associating particular access identities with respective paging profiles.

[0021] As further shown in FIG. **2**, UE priority level information may be represented by data structure **203**, which may include information associating different UEs **107** with corresponding priority levels or sets of priority levels. As noted above, in this example, respective access identities may be associated with respective priority levels. In some embodiments, other types of identifiers or indicators may be used in addition to, or in lieu of, access identities. Further, in this example, Subscription Permanent Identifiers ("SUPIs") may be used to distinguish between different UEs **107**. In some embodiments, additional or different identifiers may be used to identify different UEs **107**, such as a Globally Unique Temporary Identifier ("GUTI"), an International Mobile Subscriber Identity ("IMSI"), an International Mobile Station Equipment Identity ("IMEI"), an Mobile Directory Number ("MDN"), or the like.

[0022] In this example, as shown in data structure **203**, a first UE (e.g., associated with example identifier SUPI_A) may be associated with two different priority levels (e.g., indicated as access identities **11** and **12**), a second UE (e.g., associated with example identifier SUPI_B) may be associated with a particular priority level (e.g., indicated as access identity **13**), and so on. As discussed below, the association of a given UE **107** with one or more access identities (and/or other priority level identifiers) may indicate that such UE **107** is authorized to request and/or receive service in accordance with such access identities. For example, the particular UE **107** with the identifier SUPI_A may be authorized to receive QoS parameters associated with access identities **11** and **12** (e.g., DNN_A, NSSAI_A, and/or NSSAI_B).

[0023] As shown in FIGS. **3** and **4**, one or more NFs **301** may receive priority access information for one or more UEs **107** from UDM/UDR **303**, and may use the priority access information when providing service to such UEs **107**. For example, such NFs **301** may implement QoS parameters, queuing parameters, routing or forwarding parameters, etc. for traffic associated with a given UE **107** based on identifying one or more priority levels with which UE **107** is associated.

[0024] As shown in FIG. **3**, a particular NF **301** may request (at **302**) priority access information for a particular UE **107**. NF **301** may, for example, output such request via an SBI associated with UDM/UDR **303**, such as an Nudr interface, an Nudm interface, or some other suitable interface. NF **301** may output (at **302**) the request based on a suitable triggering event or condition, such as identifying a communication session request associated with UE **107**, receiving traffic to be forwarded to UE **107**, etc. In some embodiments, the request (at **302**) may include an identifier of UE **107**, such as a SUPI, a GUTI, an MDN, and/or some other suitable identifier. In some embodiments, the request (at **302**) may include a "GET" message or other suitable type of message.

[0025] UDM/UDR **303** may maintain UE priority level information/policies **105**, and/or information derived from or represented by UE priority level information/policies **105**. UDM/UDR **303** may also maintain other suitable subscription information associated with one or more UEs

**107**. UDM/UDR **303** may be “trusted” from the standpoint of NFs **301** of RAN **101** and/or core network **103**, inasmuch as subscription information received from UDM/UDR **303** (e.g., including UE priority level information) may be considered as valid for use by NFs **301**. For example, access to provide or update information to UDM/UDR **303** may be securely managed via one or more suitable security mechanisms.

[0026] UDM/UDR **303** may respond (at **304**) to the request with priority access information for UE **107**. For example, UDM/UDR **303** may identify one or more entries in data structure **203** (or some other suitable data structure) that correspond to UE **107**, and may indicate which access identities or other priority level identifiers are authorized for UE **107**. In some situations, UDM/UDR **303** may identify that UE **107** is not associated with any access identities or other priority level identifiers, and may indicate (at **304**) that UE **107** is not associated with any access identities or other priority level identifiers. In some embodiments, UDM/UDR **303** may output (at **304**) a “200 OK” message or other suitable type of message.

[0027] As shown in FIG. **4**, in some embodiments, one or more NFs **301** may subscribe (at **402**) to priority access information for one or more UEs **107**. For example, NF **301** may output a request or indication to UDM/UDR **303** (e.g., via an Nudr interface, an Nudm interface, or some other suitable interface) requesting automatic updates to priority access information associated with UE **107** when UDM/UDR **303** receives updates to such information (e.g., modifications to access identities or other suitable priority level identifiers with which UE **107** is associated). At some point, UDM/UDR **303** may receive (at **404**) updated priority access information for UE **107**. For example, UDM/UDR **303** may receive such information from management platform **401** or some other suitable device or system that is authorized to update or modify information stored by UDM/UDR **303**. The updated priority access information for UE **107** may, for example, add one or more access identities and/or to remove one or more access identities with which UE **107** is associated.

[0028] Based on receiving (at **404**) the updated priority access information for UE **107**, and further based on the subscribing (at **402**) of NF **301** to updated priority access information for UE **107**, UDM/UDR **303** may automatically output (at **406**) the updated priority access information to NF **301**. For example, UDM/UDR **303** may “push” the updated priority access information to NF **301**. That is, UDM/UDR **303** may output (at **406**) the updated priority access information without receiving a request from NF **301** for such information after receiving (at **404**) the updated information from management platform **401**. As discussed above, NF **301** may utilize the updated priority access information for UE **107** when providing services to UE **107**, such as traffic routing, processing, forwarding, etc. services. As discussed above, implementing the updated priority access information may include identifying and/or utilizing QoS parameters associated with the updated priority access information for UE **107**.

[0029] FIG. **5** illustrates an example of particular NFs **301** that may utilize priority access information when providing services to UE **107**. For example, as shown, Access and Mobility Management Function (“AMF”) **501** and/or Session Management Function (“SMF”) **503** may receive UE priority level information/policies **105**. As discussed above, AMF **501** and SMF **503** may receive such information based on requesting (e.g., at **302**) such information from UDM/UDR **303**, and/or may automatically receive (e.g., at **406**) such information from UDM/UDR **303** in a “push” fashion. As also shown, UE **107** may maintain UE priority level information **505**, indicating one or more priority levels with which UE **107** is associated. In some embodiments, UE **107** may maintain such information in a secure subscriber information component, such as a Subscriber Identification Module (“SIM”), a Universal Subscriber Identity Module (“USIM”), a Universal Integrated Circuit Card (“UICC”), an embedded Subscriber Identity Module (“eSIM”), or the like.

[0030] AMF **501** may perform access control for one or more UEs **107** based on UE priority level and/or other suitable factors. For example, UE **107** may request access to RAN **101**, and may include one or more access identities or other suitable priority level indicators in the request. In some embodiments, UE **107** may include the one or more access identities or other suitable priority level indicators as an Radio Resource Control ("RRC") establishment cause in one or more RRC messages sent to AMF **501** (e.g., via an N1 interface) when requesting access to RAN **101**. AMF **501** may, for example, accept or deny the access request based on the requesting priority level.

[0031] Similar concepts discussed above may be applicable when UE priority level information/policies **105**, as maintained by AMF **501**, do not include UE priority level information for UE **107**, and/or when UE priority level information/policies **105** do not indicate that UE **107** is authorized for a requested priority level. That is, in some embodiments, the absence of information indicating that a given UE **107** is associated with a given requested priority level may be used to determine that such UE **107** is authorized for such priority level.

[0032] Further, AMF **501** may communicate with one or more other elements of RAN **101**, such as a base station (e.g., a Next Generation Node B ("gNB")), a RAN controller, etc. in order to implement a requested priority level for which UE **107** is authorized. For example, AMF **501** may identify one or more QoS parameters, RAN-specific parameters (e.g., one or more 5QI values), or other suitable parameters associated with the priority level, and may indicate to the base station or other elements of RAN **101** that services provided to UE **107** should be provided in accordance with such QoS parameters. Additionally, or alternatively, AMF **501** may indicate to the base station or other elements of RAN **101** that UE **107** is associated with the requested priority level. In turn the base station or other elements of RAN **101** may implement QoS parameters or other suitable parameters, associated with the requested priority level, when providing services to UE **107**. For example, the base station may schedule or allocate radio resources for UE **107** based on the priority level, may designate one or more radio bearers for UE **107** with one or more QCI values, etc.

[0033] Additionally, as discussed above, AMF **501** may implement a particular paging profile associated with the requested priority level. For example, AMF **501** may output paging messages, perform location determination techniques, output downlink data notifications, and/or otherwise perform paging operations or other operations in accordance with the requested priority level (e.g., as indicated by UE priority level information/policies **105**). As discussed above, the different paging profiles may include paging UE **107** more or less often, setting minimum or maximum thresholds for how often UE **107** is paged, etc.

[0034] As another example, UE **107** may request the establishment of one or more communication sessions with core network **103**, such as one or more protocol data unit ("PDU") sessions. In some embodiments, such requests may be received by SMF **503** (e.g., UE **107** may output the requests to AMF **501**, which may forward such requests to SMF **503** via an N3 interface or other suitable interface). In accordance with some embodiments, the request may include an identification of one or more access identities. For example, UE **107** may have identified one or more access identities indicated in UE priority level information **505**, and may request that the communication session be associated with one or more of such access identities.

[0035] SMF **503** may compare the requested access identities to access identities indicated in UE priority level information/policies **105** as being authorized for UE **107**, and may accept or deny the request based on the comparing. In situations where UE **107** is authorized for the requested access identity, SMF **503** may perform one or more additional operations and/or may communicate with one or more other network elements (e.g., a User Plane Function ("UPF"), a Policy Control Function ("PCF"), etc.) to facilitate establishment of the requested communication session. In some embodiments, SMF **503** may instruct UPF and/or one or more other elements of core network **103** (e.g., one or more routers, hubs, switches, etc. that route traffic within core network **103**) to mark traffic, associated with the requested communication session, with one or more labels, markers, identifiers, etc. that are associated with QoS parameters indicated by UE priority level information/policies **105** as being associated with the requested access identity. In some embodiments, such markings may include Differentiated Services Code Point ("DSCP") markings or other suitable markings, based on which elements of core network **103** may implement such

QoS parameters.

[0036] FIG. **6** illustrates an example signal flow for providing subscription-based priority access to one or more UEs **107** in a wireless network. As shown, UE **107** may output (at **602**) an access request to AMF **501** (e.g., as part of a connection procedure between UE **107** and RAN **101**), which may include a requested priority level. For example, as discussed above, the request may include one or more RRC messages that include an RRC establishment cause that includes one or more requested access identities. AMF **501** may obtain (at **604**) subscription information for UE **107** from UDM/UDR **303**, which may include one or more priority levels (e.g., access identities) for which UE **107** is authorized. AMF **501** may determine (at **606**) whether UE **107** is authorized for the requested priority level based on the subscription information received from UDM/UDR **303**. AMF **501** may accept or deny (at **608**) the access request based on the determination (at **606**) of whether UE **107** is authorized for the requested priority level and/or based on other factors. In some embodiments, AMF **501** may perform one or more other operations based on determining that UE **107** is authorized for the requested priority level, such as implementing one or more corresponding paging profiles with respect to UE **107**.

[0037] Assuming the access request was accepted (at **608**), UE **107** may subsequently output (at **610**) a communication session request, such as a PDU session establishment request, to SMF **503**. In some embodiments, UE **107** may output the request to AMF **501**, which may forward the request to SMF **503**. As discussed above, the request (at **610**) may include an access identity or other suitable priority level identifier. SMF **503** may obtain (at **612**) subscription information for UE **107** from UDM/UDR **303**, which may include one or more priority levels (e.g., access identities) for which UE **107** is authorized. SMF **503** may determine (at **614**) whether UE **107** is authorized for the requested priority level based on the subscription information received from UDM/UDR **303**. SMF **503** may accept or deny (at **616**) the session establishment request based on the determination (at **614**) of whether UE **107** is authorized for the requested priority level and/or based on other factors. As discussed above, accepting the session establishment request may include communicating with one or more other elements of core network **103** to implement the requested session, such as communicating with a UPF to establish the PDU session between UE **107** and the UPF. In some embodiments, accepting the communication session may include identifying a network slice with which the priority level is associated, and selecting a UPF or other suitable devices or systems that are associated with the identified network slice.

[0038] AMF **501** and SMF **503** are presented as example NFs **301** that may request, receive, utilize, etc. priority level information for one or more UEs **107**. In some embodiments, other NFs **301** (e.g., in addition to or in lieu of AMF **501** and/or SMF **503**) may similarly obtain priority level information associated with one or more UEs **107** (e.g., from UDM/UDR **303**), and may implement policies, parameters, etc. associated with such priority levels when providing service to such UEs **107**.

[0039] FIG. **7** illustrates an example process **700** for providing subscription-based priority access to a UE in a wireless network. In some embodiments, one or more portions of process **700** may be performed by UDM/UDR **303**. In some embodiments, one or more other devices may perform some or all of process **700** in concert with, and/or in lieu of, UDM/UDR **303**. For example, one or more NFs **301** may perform one or more operations of process **700**.

[0040] As shown, process **700** may include maintaining (at **702**) subscription information associated with one or more UEs **107**, including priority level information. For example, as discussed above, UDM/UDR **303** may receive and maintain subscription information associated with one or more UEs **107** from management platform **401** and/or some other suitable authorized source. The subscription information may include parameters such as device identifiers (e.g., SUPI, GUTI, MDN, etc.), user information, location information, etc. In accordance with some embodiments, the subscription information may include one or more priority level identifiers, such as access identities (e.g., including access identities **11-15**).

[0041] Process **700** may further include providing (at **704**) subscription information, associated with a particular UE **107**, to a particular NF **301**. For example, as discussed above, NF **301** may request (e.g., at **302**) subscription information associated with the particular UE **107** (e.g., based on a triggering event such as a connection request from UE **107**), and/or may automatically output or “push” (e.g., at **406**) the subscription information to NF **301**. The provided subscription information may include one or more priority level identifiers with which the particular UE **107** is associated (e.g., is authorized to use, access, request, etc.).

[0042] Process **700** may additionally include maintaining (at **706**) priority level policy information associating priority levels with respective network parameters. For example, NF **301** may receive (e.g., from management platform **401** or some other suitable authorized source) priority level policy information (e.g., as shown in data structure **201**) associating particular priority level identifiers (e.g., access identities) with particular network parameters (e.g., QoS parameters or other suitable parameters).

[0043] Process **700** may also include providing (at **708**) service to the particular UE **107** based on the priority level information with which the particular UE **107** is associated, as well as based on the priority level policy information. For example, NF **301** may receive or identify uplink traffic for UE **107** (e.g., traffic sent by UE **107**) and/or downlink traffic (e.g., traffic sent to UE **107**), and may queue, schedule, process, forward, filter, etc. the traffic in accordance with the priority level information. For example, NF **301** may apply a particular set of QoS parameters, weights, etc. to the traffic based on the priority level information. In this manner, different priority levels may be able to be provided on a per-UE basis, thus enhancing operator control of the network as well as enhancing the user experience of users of UEs **107**. Additionally, security is maintained by virtue of subscription information being maintained by a secure network element (e.g., UDM/UDR **303**), where such subscription information includes priority level information. In this sense, requests from UEs **107** for access to a particular priority level may be verified by checking the subscription information maintained by the secure network element.

[0044] FIG. **8** illustrates an example environment **800**, in which one or more embodiments may be implemented. In some embodiments, environment **800** may correspond to a Fifth Generation (“5G”) network, and/or may include elements of a 5G network. In some embodiments, environment **800** may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G radio access technology (“RAT”) may be used in conjunction with one or more other RATs (e.g., a Long-Term Evolution (“LTE”) RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”)). In some embodiments, portions of environment **800** may represent or may include a 5G core (“5GC”). As shown, environment **800** may include UE **107**, RAN **810** (which may include one or more gNBs **811**), RAN **812** (which may include one or more evolved Node Bs (“eNBs”) **813**), and various network functions such as AMF **501**, Mobility Management Entity (“MME”) **816**, Serving Gateway (“SGW”) **817**, SMF/Packet Data Network (“PDN”) Gateway (“PGW”)-Control plane function (“PGW-C”) **820**, PCF/Policy Charging and Rules Function (“PCRF”) **825**, Application Function (“AF”) **830**, UPF/PGW-User plane function (“PGW-U”) **835**, UDM/Home Subscriber Server (“HSS”) **840**, Authentication Server Function (“AUSF”) **845**, and Network Exposure Function (“NEF”)/Service Capability Exposure Function (“SCEF”) **849**. Environment **800** may also include one or more networks, such as Data Network (“DN”) **850**. Environment **800** may include one or more additional devices or systems communicatively coupled to one or more networks (e.g., DN **850**), such as one or more external devices **854**.

[0045] The example shown in FIG. **8** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **820**, PCF/PCRF **825**, UPF/PGW-U **835**, UDM/HSS **840**, and/or AUSF **845**). In practice, environment **800** may include multiple instances of such components or functions. For example, in some embodiments, environment **800** may include multiple “slices” of a core network, where each slice includes a discrete and/or logical set of network functions (e.g., one slice may include a first instance of AMF **501**, SMF/PGW-C **820**, PCF/PCRF **825**, and/or UPF/PGW-U **835**, while another slice may include a second instance of AMF **501**, SMF/PGW-C **820**, PCF/PCRF **825**, and/or UPF/PGW-U **835**). The different slices may provide differentiated levels of service, such as service in accordance with different QoS parameters.

[0046] The quantity of devices and/or networks, illustrated in FIG. **8**, is provided for explanatory purposes only. In practice, environment **800** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **8**. For example, while not shown, environment **800** may include devices that facilitate or enable communication between various components shown in environment **800**, such as routers, modems, gateways, switches, hubs, etc. In some implementations, one or more devices of environment **800** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **800**. Alternatively, or additionally, one or more of the devices of environment **800** may perform one or more network functions described as being performed by another one or more of the devices of environment **800**.

[0047] Additionally, one or more elements of environment **800** may be implemented in a virtualized and/or containerized manner. For example, one or more of the elements of environment **800** may be implemented by one or more Virtualized NFs (“VNFs”), Cloud-Native NFs (“CNFs”), etc. In such embodiments, environment **800** may include, may implement, and/or may be communicatively coupled to an orchestration platform that provisions hardware resources, installs containers or applications, performs load balancing, and/or otherwise manages the deployment of such elements of environment **800**. In some embodiments, such orchestration and/or management of such elements of environment **800** may be performed by, or in conjunction with, the open-source Kubernetes® application programming interface (“API”) or some other suitable virtualization, containerization, and/or orchestration system.

[0048] Elements of environment **800** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. Examples of interfaces or communication pathways between the elements of environment **800**, as shown in FIG. **8**, may include an N1 interface, an N2 interface, an N3 interface, an N4 interface, an N5 interface, an N6 interface, an N7 interface, an N8 interface, an N9 interface, an N10 interface, an N11 interface, an N12 interface, an N13 interface, an N14 interface, an N15 interface, an N26 interface, an S1-C interface, an S1-U interface, an S5-C interface, an S5-U interface, an S6a interface, an S11 interface, and/or one or more other interfaces. Such interfaces may include interfaces not explicitly shown in FIG. **8**, such as Service-Based Interfaces (“SBIs”), including an Namf interface, an Nudm interface, an Npcf interface, an Nupf interface, an Nnef interface, an Nsmf interface, and/or one or more other SBIs. In some embodiments, environment **800** may be, may include, may be implemented by, and/or may be communicatively coupled to RAN **101** and/or core network **103**.

[0049] UE **107** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **810**, RAN **812**, and/or DN **850**. UE **107** may be, or may include, a radiotelephone, a personal communications system (“PCS”) terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant (“PDA”) (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an IoT device (e.g., a sensor, a smart home appliance, a wearable device, a programmable logic controller or other industrial controller, a Machine-to-Machine (“M2M”) device, or the like), a Fixed Wireless Access (“FWA”) device, or another type of mobile computation and communication device. UE **107** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **850** via RAN **810**, RAN **812**, and/or UPF/PGW-U

**835**.

[0050] RAN **810** may be, or may include, a 5G RAN that implements a 5G RAT and that includes one or more base stations (e.g., one or more gNBs **811**), via which UE **107** may communicate with one or more other elements of environment **800**. UE **107** may communicate with RAN **810** via an air interface (e.g., as provided by gNB **811**). For instance, RAN **810** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, etc.) from UE **107** via the air interface, and may communicate the traffic to UPF/PGW-U **835** and/or one or more other devices or networks. Further, RAN **810** may receive signaling traffic, control plane traffic, etc. from UE **107** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to AMF **501** and/or one or more other devices or networks. Additionally, RAN **810** may receive traffic intended for UE **107** (e.g., from UPF/PGW-U **835**, AMF **501**, and/or one or more other devices or networks) and may communicate the traffic to UE **107** via the air interface. In some embodiments, RAN **101** may implement, may be implemented by, and/or may otherwise be associated with RAN **810**.

[0051] RAN **812** may be, or may include, an LTE RAN that implements an LTE RAT and that includes one or more base stations (e.g., one or more eNBs **813**), via which UE **107** may communicate with one or more other elements of environment **800**. UE **107** may communicate with RAN **812** via an air interface (e.g., as provided by eNB **813**). For instance, RAN **812** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **107** via the air interface, and may communicate the traffic to UPF/PGW-U **835** (e.g., via SGW **817**) and/or one or more other devices or networks. Further, RAN **812** may receive signaling traffic, control plane traffic, etc. from UE **107** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to MME **816** and/or one or more other devices or networks. Additionally, RAN **812** may receive traffic intended for UE **107** (e.g., from UPF/PGW-U **835**, MME **816**, SGW **817**, and/or one or more other devices or networks) and may communicate the traffic to UE **107** via the air interface. In some embodiments, RAN **101** may implement, may be implemented by, and/or may otherwise be associated with RAN **812**.

[0052] One or more RANs of environment **800** (e.g., RAN **810** and/or RAN **812**) may include, may implement, and/or may otherwise be communicatively coupled to one or more edge computing devices, such as one or more Multi-Access/Mobile Edge Computing ("MEC") devices (referred to sometimes herein simply as a "MECs") **814**. MECs **814** may be co-located with wireless network infrastructure equipment of RANs **810** and/or **812** (e.g., one or more gNBs **811** and/or one or more eNBs **813**, respectively). Additionally, or alternatively, MECs **814** may otherwise be associated with geographical regions (e.g., coverage areas) of wireless network infrastructure equipment of RANs **810** and/or **812**. In some embodiments, one or more MECs **814** may be implemented by the same set of hardware resources, the same set of devices, etc. that implement wireless network infrastructure equipment of RANs **810** and/or **812**. In some embodiments, one or more MECs **814** may be implemented by different hardware resources, a different set of devices, etc. from hardware resources or devices that implement wireless network infrastructure equipment of RANs **810** and/or **812**. In some embodiments, MECs **814** may be communicatively coupled to wireless network infrastructure equipment of RANs **810** and/or **812** (e.g., via a high-speed and/or low-latency link such as a physical wired interface, a high-speed and/or low-latency wireless interface, or some other suitable communication pathway).

[0053] MECs **814** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **107**, via RAN **810** and/or **812**. For example, RAN **810** and/or **812** may route some traffic from UE **107** (e.g., traffic associated with one or more particular services, applications, application types, etc.) to a respective MEC **814** instead of to core network elements of **800** (e.g., UPF/PGW-U **835**). MEC **814** may accordingly provide services to UE **107** by processing such traffic, performing one or more computations based on the received traffic, and providing traffic to UE **107** via RAN

**810** and/or **812**. MEC **814** may include, and/or may implement, some or all of the functionality described above with respect to UPF/PGW-U **835**, AF **830**, one or more application servers, and/or one or more other devices, systems, VNFs, CNFs, etc. In this manner, ultra-low latency services may be provided to UE **107**, as traffic does not need to traverse links (e.g., backhaul links) between RAN **810** and/or **812** and the core network.

[0054] AMF **501** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **107** with the 5G network, to establish bearer channels associated with a session with UE **107**, to hand off UE **107** from the 5G network to another network, to hand off UE **107** from the other network to the 5G network, manage mobility of UE **107** between RANs **810** and/or gNBs **811**, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs **501**, which communicate with each other via the N14 interface (denoted in FIG. **8** by the line marked “N14” originating and terminating at AMF **501**).

[0055] MME **816** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **107** with the EPC, to establish bearer channels associated with a session with UE **107**, to hand off UE **107** from the EPC to another network, to hand off UE **107** from another network to the EPC, manage mobility of UE **107** between RANs **812** and/or eNBs **813**, and/or to perform other operations.

[0056] SGW **817** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate traffic received from one or more eNBs **813** and send the aggregated traffic to an external network or device via UPF/PGW-U **835**. Additionally, SGW **817** may aggregate traffic received from one or more UPF/PGW-Us **835** and may send the aggregated traffic to one or more eNBs **813**. SGW **817** may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs **810** and **812**).

[0057] SMF/PGW-C **820** may include one or more devices, systems, VNFs, CNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C **820** may, for example, facilitate the establishment of communication sessions on behalf of UE **107**. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF **825**.

[0058] PCF/PCRF **825** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF **825** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF **825**).

[0059] AF **830** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

[0060] UPF/PGW-U **835** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U **835** may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE **107**, from DN **850**, and may forward the user plane data toward UE **107** (e.g., via RAN **810**, SMF/PGW-C **820**, and/or one or more other devices). In some embodiments, multiple instances of UPF/PGW-U **835** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **107** may be coordinated via the N9 interface (e.g., as denoted in FIG. **8** by the line marked “N9” originating and terminating at UPF/PGW-U **835**). Similarly, UPF/PGW-U **835** may receive traffic from UE **107** (e.g., via RAN **810**, RAN **812**, SMF/PGW-C **820**, and/or one or more other devices), and may forward the traffic toward DN **850**. In some embodiments, UPF/PGW-U **835** may communicate (e.g., via the N4 interface) with SMF/PGW-C **820**, regarding user plane data processed by UPF/PGW-U **835**.

[0061] UDM/HSS **840** and AUSF **845** may include one or more devices, systems, VNFs, CNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF **845** and/or UDM/HSS **840**, profile information associated with a subscriber. In some embodiments, UDM/HSS **840** may include, may implement, may be communicatively coupled to, and/or may otherwise be associated with some other type of repository or database, such as a UDR. For example, UDM/UDR **303** may perform some or all of the functions of UDM/HSS **840**, and/or UDM/HSS **840** may perform some or all of the functions of UDM/UDR **303**. AUSF **845** and/or UDM/HSS **840** may perform authentication, authorization, and/or accounting operations associated with one or more UEs **107** and/or one or more communication sessions associated with one or more UEs **107**.

[0062] DN **850** may include one or more wired and/or wireless networks. For example, DN **850** may include an Internet Protocol ("IP")-based PDN, a wide area network ("WAN") such as the Internet, a private enterprise network, and/or one or more other networks. UE **107** may communicate, through DN **850**, with data servers, other UEs **107**, and/or to other servers or applications that are coupled to DN **850**. DN **850** may be connected to one or more other networks, such as a public switched telephone network ("PSTN"), a public land mobile network ("PLMN"), and/or another network. DN **850** may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE **107** may communicate.

[0063] External devices **854** may include one or more devices or systems that communicate with UE **107** via DN **850** and one or more elements of **800** (e.g., via UPF/PGW-U **835**). In some embodiments, external devices **854** may include, may implement, and/or may otherwise be associated with management platform **401**. External devices **854** may include, for example, one or more application servers, content provider systems, web servers, or the like. External devices **854** may, for example, implement "server-side" applications that communicate with "client-side" applications executed by UE **107**. External devices **854** may provide services to UE **107** such as gaming services, videoconferencing services, messaging services, email services, web services, and/or other types of services.

[0064] In some embodiments, external devices **854** may communicate with one or more elements of environment **800** (e.g., core network elements) via NEF/SCEF **849**. NEF/SCEF **849** include one or more devices, systems, VNFs, CNFs, etc. that provide access to information, APIs, and/or other operations or mechanisms of one or more core network elements to devices or systems that are external to the core network (e.g., to external device **854** via DN **850**). NEF/SCEF **849** may maintain authorization and/or authentication information associated with such external devices or systems, such that NEF/SCEF **849** is able to provide information, that is authorized to be provided, to the external devices or systems. For example, a given external device **854** may request particular information associated with one or more core network elements. NEF/SCEF **849** may authenticate the request and/or otherwise verify that external device **854** is authorized to receive the information, and may request, obtain, or otherwise receive the information from the one or more core network elements. In some embodiments, NEF/SCEF **849** may include, may implement, may be implemented by, may be communicatively coupled to, and/or may otherwise be associated with a Security Edge Protection Proxy ("SEPP"), which may perform some or all of the functions discussed above. External device **854** may, in some situations, subscribe to particular types of requested information provided by the one or more core network elements, and the one or more core network elements may provide (e.g., "push") the requested information to NEF/SCEF **849** (e.g., in a periodic or otherwise ongoing basis).

[0065] In some embodiments, external devices **854** may communicate with one or more elements of RAN **810** and/or **812** via an API or other suitable interface. For example, a given external device **854** may provide instructions, requests, etc. to RAN **810** and/or **812** to provide one or more services via one or more respective MECs **814**. In some embodiments, such instructions, requests, etc. may include QoS parameters, Service Level Agreements ("SLAs"), etc. (e.g., maximum latency thresholds, minimum throughput thresholds, etc.) associated with the services.

[0066] FIG. **9** illustrates another example environment **900**, in which one or more embodiments may be implemented. In some embodiments, environment **900** may correspond to a 5G network, and/or may include elements of a 5G network. In some embodiments, environment **900** may correspond to a 5G SA architecture. In some embodiments, environment **900** may include a 5GC, in which 5GC network elements perform one or more operations described herein.

[0067] As shown, environment **900** may include UE **107**, RAN **810** (which may include one or more gNBs **811** or other types of wireless network infrastructure) and various network functions, which may be implemented as VNFs, CNFs, etc. Such network functions may include AMF **501**, SMF **503**, UPF **905**, PCF **907**, UDM **909**, AUSF **845**, Network Repository Function ("NRF") **911**, AF **830**, UDR **913**, and NEF **915**. Environment **900** may also include or may be communicatively coupled to one or more networks, such as DN **850**.

[0068] The example shown in FIG. **9** illustrates one instance of each network component or function (e.g., one instance of SMF **503**, UPF **905**, PCF **907**, UDM **909**, AUSF **845**, etc.). In practice, environment **900** may include multiple instances of such components or functions. For example, in some embodiments, environment **900** may include multiple "slices" of a core network, where each slice includes a discrete and/or logical set of network functions (e.g., one slice may include a first instance of SMF **503**, PCF **907**, UPF **905**, etc., while another slice may include a second instance of SMF **503**, PCF **907**, UPF **905**, etc.). Additionally, or alternatively, one or more of the network functions of environment **900** may implement multiple network slices. The different slices may provide differentiated levels of service, such as service in accordance with different QoS parameters.

[0069] The quantity of devices and/or networks, illustrated in FIG. **9**, is provided for explanatory purposes only. In practice, environment **900** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **9**. For example, while not shown, environment **900** may include devices that facilitate or enable communication between various components shown in environment **900**, such as routers, modems, gateways, switches, hubs, etc. In some implementations, one or more devices of environment **900** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **900**. Alternatively, or additionally, one or more of the devices of environment **900** may perform one or more network functions described as being performed by another one or more of the devices of environment **900**.

[0070] Elements of environment **900** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. Examples of interfaces or communication pathways between the elements of environment **900**, as shown in FIG. **9**, may include interfaces shown in FIG. **9** and/or one or more interfaces not explicitly shown in FIG. **9**. These interfaces may include interfaces between specific network functions, such as an N1 interface, an N2 interface, an N3 interface, an N6 interface, an N9 interface, an N14 interface, an N16 interface, and/or one or more other interfaces. In some embodiments, one or more elements of environment **900** may communicate via a service-based architecture ("SBA"), in which a routing mesh or other suitable routing mechanism may route communications to particular network functions based on interfaces or identifiers associated with such network functions. Such interfaces may include or may be referred to as SBIs, including an Namf interface (e.g., indicating communications to be routed to AMF **501**), an Nudm interface (e.g., indicating communications to be routed to UDM **909**), an Npcf interface, an Nupf interface, an Nnef interface, an Nsmf interface, an Nnrf interface, an Nudr interface, an Naf interface, and/or one or more other SBIs. In some embodiments, environment **900** may be, may include, may be implemented by, and/or may be communicatively coupled to RAN **101** and/or core network **103**.

[0071] UPF **905** may include one or more devices, systems, VNFs, CNFs, etc., that receive, route, process, and/or forward traffic (e.g., user plane traffic). As discussed above, UPF **905** may communicate with UE **107** via one or more communication sessions, such as PDU sessions. Such PDU sessions may be associated with a particular network slice or other suitable QoS parameters, as noted above. UPF **905** may receive downlink user plane traffic (e.g., voice call traffic, data traffic, etc. destined for UE **107**) from DN **850**, and may forward the downlink user plane traffic toward UE **107** (e.g., via RAN **810**). In some embodiments, multiple UPFs **905** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **107** may be coordinated via the N9 interface. Similarly, UPF **905** may receive uplink traffic from UE **107** (e.g., via RAN **810**), and may forward the traffic toward DN **850**. In some embodiments, UPF **905** may implement, may be implemented by, may be communicatively coupled to, and/or may otherwise be associated with UPF/PGW-U **835**. In some embodiments, UPF **905** may communicate (e.g., via the N4 interface) with SMF **503**, regarding user plane data processed by UPF **905** (e.g., to provide analytics or reporting information, to receive policy and/or authorization information, etc.).

[0072] PCF **907** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate, derive, generate, etc. policy information associated with the 5GC and/or UEs **107** that communicate via the 5GC and/or RAN **810**. PCF **907** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases (e.g., UDM **909**, UDR **913**, etc.), and/or from one or more users such as, for example, an administrator associated with PCF **907**. In some embodiments, the functionality of PCF **907** may be split into multiple network functions or subsystems, such as access and mobility PCF (“AM-PCF”) **917**, session management PCF (“SM-PCF”) **919**, UE PCF (“UE-PCF”) **921**, and so on. Such different “split” PCFs may be associated with respective SBIs (e.g., AM-PCF **917** may be associated with an Nampcf SBI, SM-PCF **919** may be associated with an Nsmpcf SBI, UE-PCF **921** may be associated with an Nuepcf SBI, and so on) via which other network functions may communicate with the split PCFs. The split PCFs may maintain information regarding policies associated with different devices, systems, and/or network functions.

[0073] NRF **911** may include one or more devices, systems, VNFs, CNFs, etc. that maintain routing and/or network topology information associated with the 5GC. For example, NRF **911** may maintain and/or provide IP addresses of one or more network functions, routes associated with one or more network functions, discovery and/or mapping information associated with particular network functions or network function instances (e.g., whereby such discovery and/or mapping information may facilitate the SBA), and/or other suitable information.

[0074] UDR **913** may include one or more devices, systems, VNFs, CNFs, etc. that provide user and/or subscriber information, based on which PCF **907** and/or other elements of environment **900** may determine access policies, QoS policies, charging policies, or the like. In some embodiments, UDR **913** may receive such information from UDM **909** and/or one or more other sources. In some embodiments, UDM/UDR **303** may include, may implement, may be implemented by, may be communicatively coupled to, and/or may otherwise be associated with UDM **909** and/or UDR **913**.

[0075] NEF **915** include one or more devices, systems, VNFs, CNFs, etc. that provide access to information, APIs, and/or other operations or mechanisms of the 5GC to devices or systems that are external to the 5GC. NEF **915** may maintain authorization and/or authentication information associated with such external devices or systems, such that NEF **915** is able to provide information, that is authorized to be provided, to the external devices or systems. Such information may be received from other network functions of the 5GC (e.g., as authorized by an administrator or other suitable entity associated with the 5GC), such as SMF **503**, UPF **905**, a charging function (“CHF”) of the 5GC, and/or other suitable network function. NEF **915** may communicate with external devices or systems (e.g., external devices **854**) via DN **850** and/or other suitable communication pathways.

[0076] While environment **900** is described in the context of a 5GC, as noted above, environment **900** may, in some embodiments, include or implement one or more other types of core networks. For example, in some embodiments, environment **900** may be or may include a converged packet core, in which one or more elements may perform some or all of the functionality of one or more 5GC network functions and/or one or more EPC network functions. For example, in some embodiments, AMF **501** may include, may implement, may be implemented by, and/or may otherwise be associated with MME **816**; SMF **503** may include, may implement, may be implemented by, and/or may otherwise be associated with SGW **817**; PCF **907** may include, may implement, may be implemented by, and/or may otherwise be associated with a PCRF (e.g., PCF/PCRF **825**); NEF **915** may include, may implement, may be implemented by, and/or may otherwise be associated with a SCEF (e.g., NEF/SCEF **849**); and so on.

[0077] FIG. **10** illustrates an example RAN environment **1000**, which may be included in and/or implemented by one or more RANs (e.g., RAN **810** or some other RAN). In some embodiments, a particular RAN **810** may include one RAN environment **1000**. In some embodiments, a particular RAN **810** may include multiple RAN environments **1000**. In some embodiments, RAN environment **1000** may correspond to a particular gNB **811** of RAN **810**. In some embodiments, RAN environment **1000** may correspond to multiple gNBs **811**. In some embodiments, RAN environment **1000** may correspond to one or more other types of base stations of one or more other types of RANs. As shown, RAN environment **1000** may include Central Unit ("CU") **1005**, one or more Distributed Units ("DUs") **1003-1** through **1003**-M (referred to individually as "DU **1003**," or collectively as "DUs **1003**"), and one or more Radio Units ("RUs") **1001-1** through **1001**-M (referred to individually as "RU **1001**," or collectively as "RUs **1001**").

[0078] CU **1005** may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. **9**, such as AMF **501** and/or UPF **905**) and/or some other device or system such as MEC **814**. In the uplink direction (e.g., for traffic from UEs **107** to a core network), CU **1005** may aggregate traffic from DUs **1003**, and forward the aggregated traffic to the core network. In some embodiments, CU **1005** may receive traffic according to a given protocol (e.g., Radio Link Control ("RLC") traffic) from DUs **1003**, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol ("PDCP") packets based on the RLC packets) on the traffic received from DUs **1003**.

[0079] CU **1005** may receive downlink traffic (e.g., traffic from the core network, traffic from a given MEC **814**, etc.) for a particular UE **107**, and may determine which DU(s) **1003** should receive the downlink traffic. DU **1003** may include one or more devices that transmit traffic between a core network (e.g., via CU **1005**) and UE **107** (e.g., via a respective RU **1001**). DU **1003** may, for example, receive traffic from RU **1001** at a first layer (e.g., physical ("PHY") layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or RLC). DU **1003** may receive traffic from CU **1005** at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU **1001** for transmission to UE **107**.

[0080] RU **1001** may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs **107**, one or more other DUs **1003** (e.g., via RUs **1001** associated with DUs **1003**), and/or any other suitable type of device. In the uplink direction, RU **1001** may receive traffic from UE **107** and/or another DU **1003** via the RF interface and may provide the traffic to DU **1003**. In the downlink direction, RU **1001** may receive traffic from DU **1003**, and may provide the traffic to UE **107** and/or another DU **1003**.

[0081] One or more elements of RAN environment **1000** may, in some embodiments, be communicatively coupled to one or more MECs **814**. For example, DU **1003-1** may be communicatively coupled to MEC **814-1**, DU **1003**-M may be communicatively coupled to MEC **814**-N, CU **1005** may be communicatively coupled to MEC **814-2**, and so on. MECs **814** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **107**, via a respective RU **1001**.

[0082] For example, DU **1003-1** may route some traffic, from UE **107**, to MEC **814-1** instead of to a core network via CU **1005**. MEC **814-1** may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE **107** via RU **1001-1**. As discussed above, MEC **814** may include, and/or may implement, some or all of the functionality described above with respect to UPF **905**, AF **830**, and/or one or more other devices, systems, VNFs, CNFs, etc. In this manner, ultra-low latency services may be provided to UE **107**, as traffic does not need to traverse DU **1003**, CU **1005**, links between DU **1003** and CU **1005**, and an intervening backhaul network between RAN environment **1000** and the core network.

[0083] FIG. **11** illustrates example components of device **1100**. One or more of the devices described above may include one or more devices **1100**. Device **1100** may include bus **1110**, processor **1120**, memory **1130**, input component **1140**, output component **1150**, and communication interface **1160**. In another implementation, device **1100** may include additional, fewer, different, or differently arranged components.

[0084] Bus **1110** may include one or more communication paths that permit communication among the components of device **1100**. Processor **1120** may include a processor, microprocessor, a set of provisioned hardware resources of a cloud computing system, or other suitable type of hardware that interprets and/or executes instructions (e.g., processor-executable instructions). In some embodiments, processor **1120** may be or may include one or more hardware processors. Memory **1130** may include any type of dynamic storage device that may store information and instructions for execution by processor **1120**, and/or any type of non-volatile storage device that may store information for use by processor **1120**.

[0085] Input component **1140** may include a mechanism that permits an operator to input information to device **1100** and/or other receives or detects input from a source external to input component **1140**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **1140** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System (“GPS”)-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **1150** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes (“LEDs”), etc.

[0086] Communication interface **1160** may include any transceiver-like mechanism that enables device **1100** to communicate with other devices and/or systems (e.g., via RAN **810**, RAN **812**, DN **850**, etc.). For example, communication interface **1160** may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface **1160** may include a wireless communication device, such as an infrared (“IR”) receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a cellular radio, a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **1100** may include more than one communication interface **1160**. For instance, device **1100** may include an optical interface, a wireless interface, an Ethernet interface, and/or one or more other interfaces.

[0087] Device **1100** may perform certain operations relating to one or more processes described above. Device **1100** may perform these operations in response to processor **1120** executing instructions, such as software instructions, processor-executable instructions, etc. stored in a computer-readable medium, such as memory **1130**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The instructions may be read into memory **1130** from another computer-readable medium or from another device. The instructions stored in memory **1130** may be processor-executable instructions that cause processor **1120** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0088] The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

[0089] For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. **1-7**), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

[0090] The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

[0091] In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

[0092] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

[0093] Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

[0094] To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

[0095] No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

## Claims

**1**. A device, comprising: one or more processors configured to: maintain subscription information associated with a plurality of User Equipment (“UEs”), wherein the subscription information includes priority level information associating one or more UEs with one or more respective priority levels; receive, from a network function (“NF”) of a wireless network, a request for priority level information associated with a particular UE; identify a particular set of priority levels indicated by the subscription information as being associated with the particular UE; and output, to the NF, the identified particular set of priority levels, wherein the NF maintains policy information associating the particular set of priority levels with one or more network parameters and implements the one or more network parameters when providing service associated with the particular UE.
**2**. The device of claim 1, wherein implementing the one or more network parameters includes: identifying a particular set of Quality of Service (“QoS”) parameters with which a particular priority level, of the particular set of priority levels, is associated; identifying particular traffic that is sent to or by the particular UE; and implementing the particular set of QoS parameters with respect to the particular traffic sent to or by the particular UE.
**3**. The device of claim 1, wherein the particular set of priority levels includes one or more access identities.
**4**. The device of claim 3, wherein the subscription information includes information associating one or more UE identifiers with one or more respective access identities.
**5**. The device of claim 4, wherein the one or more UE identifiers include a particular Subscription Permanent Identifier (“SUPI”) associated with the particular UE.
**6**. The device of claim 1, wherein the subscription information is maintained by at least one of: a Unified Data Management function (“UDM”), or a Unified Data Repository (“UDR”).
**7**. The device of claim 1, wherein the NF includes an Access and Mobility Management Function (“AMF”), wherein the one or more network parameters include a particular paging profile that is associated with a particular priority level of the particular set of priority levels, wherein the AMF performs paging operations with respect to the particular UE based on the particular paging profile.
**8**. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions to: maintain subscription information associated with a plurality of User Equipment (“UEs”), wherein the subscription information includes priority level information associating one or more UEs with one or more respective priority levels; receive, from a network function (“NF”) of a wireless network, a request for priority level information associated with a particular UE; identify a particular set of priority levels indicated by the subscription information as being associated with the particular UE; and output, to the NF, the identified particular set of priority levels, wherein the NF maintains policy information associating the particular set of priority levels with one or more network parameters and implements the one or more network parameters when providing service associated with the particular UE.
**9**. The non-transitory computer-readable medium of claim 8, wherein implementing the one or more network parameters includes: identifying a particular set of Quality of Service (“QoS”) parameters with which a particular priority level, of the particular set of priority levels, is associated; identifying particular traffic that is sent to or by the particular UE; and implementing the particular set of QoS parameters with respect to the particular traffic sent to or by the particular UE.
**10**. The non-transitory computer-readable medium of claim 8, wherein the particular set of priority

levels includes one or more access identities.

**11**. The non-transitory computer-readable medium of claim 10, wherein the subscription information includes information associating one or more UE identifiers with one or more respective access identities.

**12**. The non-transitory computer-readable medium of claim 11, wherein the one or more UE identifiers include a particular Subscription Permanent Identifier ("SUPI") associated with the particular UE.

**13**. The non-transitory computer-readable medium of claim 8, wherein the subscription information is maintained by at least one of: a Unified Data Management function ("UDM"), or a Unified Data Repository ("UDR").

**14**. The non-transitory computer-readable medium of claim 8, wherein the NF includes an Access and Mobility Management Function ("AMF"), wherein the one or more network parameters include a particular paging profile that is associated with a particular priority level of the particular set of priority levels, wherein the AMF performs paging operations with respect to the particular UE based on the particular paging profile.

**15**. A method, comprising: maintaining subscription information associated with a plurality of User Equipment ("UEs"), wherein the subscription information includes priority level information associating one or more UEs with one or more respective priority levels; receiving, from a network function ("NF") of a wireless network, a request for priority level information associated with a particular UE; identifying a particular set of priority levels indicated by the subscription information as being associated with the particular UE; and outputting, to the NF, the identified particular set of priority levels, wherein the NF maintains policy information associating the particular set of priority levels with one or more network parameters and implements the one or more network parameters when providing service associated with the particular UE.

**16**. The method of claim 15, wherein implementing the one or more network parameters includes: identifying a particular set of Quality of Service ("QoS") parameters with which a particular priority level, of the particular set of priority levels, is associated; identifying particular traffic that is sent to or by the particular UE; and implementing the particular set of QoS parameters with respect to the particular traffic sent to or by the particular UE.

**17**. The method of claim 15, wherein the particular set of priority levels includes one or more access identities.

**18**. The method of claim 17, wherein the subscription information includes information associating one or more UE identifiers with one or more respective access identities, and wherein the one or more UE identifiers include a particular Subscription Permanent Identifier ("SUPI") associated with the particular UE.

**19**. The method of claim 15, wherein the subscription information is maintained by at least one of: a Unified Data Management function ("UDM"), or a Unified Data Repository ("UDR").

**20**. The method of claim 15, wherein the NF includes an Access and Mobility Management Function ("AMF"), wherein the one or more network parameters include a particular paging profile that is associated with a particular priority level of the particular set of priority levels, wherein the AMF performs paging operations with respect to the particular UE based on the particular paging profile.